

Set up Python and VS Code for MAR 6930

Marketing Data Visualization · Lecture 1 · a step-by-step guide for Windows and macOS

You will install four free things: VS Code, Python, a few VS Code extensions, and two Python packages (pandas and Matplotlib). Then you will run the Lecture 1 notebook from top to bottom. You are finished when every cell runs without an error and the charts appear. Work through the steps in order; each step assumes the one before it worked.

Step 1. Install VS Code

1. Go to code.visualstudio.com/download and download the version for your computer.
2. **Windows:** run the installer and accept the default options.
macOS: open the downloaded file and drag **Visual Studio Code** into your **Applications** folder.
3. Open VS Code once to make sure it starts, then close it.

Step 2. Install Python

VS Code does not include Python, and installing its Python extension does not install Python either. Skip this step if you already have Python 3 installed, for example through Anaconda; you will choose that Python in Step 5.

Windows

1. Go to python.org/downloads and download the **Python install manager**. (The same program is in the Microsoft Store as "Python Install Manager".)
2. Open the downloaded file and click **Install**.
3. Open **Terminal** from the Start menu and run the command below. It downloads and installs the latest Python. If it asks whether to add a directory to your PATH, type y and press Enter.

```
py install default
```
4. Close Terminal, open it again, and check the installation:

```
python --version
```

You should see a version number such as Python 3.14.7.

macOS

1. Go to python.org/downloads and click the download button for the latest Python 3 release.
2. Open the downloaded .pkg file and follow the installer with the default options.
3. Open **Terminal** (Applications › Utilities › Terminal) and check the installation:

```
python3 --version
```

You should see a version number such as Python 3.14.7. On a Mac, the command is `python3`, not `python`.

Step 3. Install the VS Code extensions

1. Open VS Code and click the **Extensions** icon (four squares) in the left sidebar, or press `Ctrl+Shift+X` (Windows) / `Cmd+Shift+X` (macOS).
2. Type the ID from the table into the search box at the top of the Extensions panel.
3. Check that the publisher matches, then click **Install**. If an extension is already installed, make sure it is enabled.

Extension (publisher)	Search for this ID	Needed?
Python extension (Microsoft)	<code>ms-python.python</code>	Required
Jupyter (Microsoft)	<code>ms-toolsai.jupyter</code>	Required
Jupyter Notebook Renderers (Microsoft)	<code>ms-toolsai.jupyter-renderers</code>	Recommended: displays outputs, including charts
GitHub Copilot Chat (GitHub)	GitHub Copilot Chat	Recommended: see Step 8
Claude Code (Anthropic) or Codex (OpenAI)	Claude Code, or <code>openai.chatgpt</code>	Optional, if you already pay for one

You still need the Python extension after Step 2. Step 2 installed Python, the program that runs your code. The Python extension is a VS Code plug-in that lets VS Code find and use that Python.

Step 4. Put the course files in one folder

Download `lecture1.ipynb` and `lecture1.csv` and save them in the **same folder**, for example a folder called `MAR6930` in your Documents. The notebook reads the CSV from its own folder, so the two files must stay together.

Step 5. Open the folder and select the kernel

VS Code shows the notebook, but the code runs in a **kernel**: a Python program running in the background. In this step you tell VS Code which Python to use as the kernel.

1. In VS Code, choose **File > Open Folder** and select the folder from Step 4. Open the folder, not just the notebook file.

2. If VS Code asks whether you trust the authors of the files in this folder, click **Yes, I trust the authors**. Otherwise VS Code opens the folder in Restricted Mode, where notebooks cannot run.
3. In the Explorer panel on the left, click `lecture1.ipynb`.
4. Click **Select Kernel** at the top right of the notebook, choose **Python Environments**, and pick the Python you installed in Step 2 (or your Anaconda Python).
5. If VS Code offers to install `ipykernel`, click **Install**. This is the small package that lets Python act as a notebook kernel.

Your Python is not in the list? Quit VS Code completely and open it again. VS Code looks for Python installations when it starts.

Step 6. Install pandas and Matplotlib

A new Python installation does not include these packages, so you install them once, from inside the notebook. Installing from the notebook puts them into the kernel's Python, which is the one that needs them.

1. Scroll to section **0) Check your setup**. The first code cell contains:

```
# %pip install pandas matplotlib
```
2. Delete the `#` and the space after it, so the line reads `%pip install pandas matplotlib`.
3. Click the Run button (the triangle) to the left of the cell and wait. The output should end with `Successfully installed ... or Requirement already satisfied`.
4. Click **Restart** in the notebook toolbar to restart the kernel, so it picks up the new packages.

You only need to do this once for each Python installation. You can put the `#` back afterwards.

Step 7. Run the whole notebook

Click **Run All** in the notebook toolbar. The cells run in order, and each result appears below its cell. Scroll through and compare with the checks below.

Check: the setup cell prints `Setup works. You are ready to load the data.` The `df.info()` cell reports 448 entries and 6 columns. Two weekly line charts (revenue and margin rate) appear in section 3, a revenue chart with "Promo" labels follows, and a daily revenue chart appears in section 4.

If all of that appears with no red error messages, your setup works. Save the notebook with `Ctrl+S` (Windows) / `Cmd+S` (macOS).

Step 8. Set up GitHub Copilot (recommended)

Copilot is an AI assistant inside VS Code. Students get it free through GitHub Education. You do not need it to run the notebook, but it helps you understand code and error messages.

1. Create a GitHub account if you do not have one, then apply to [GitHub Education](#) as a student using your **ufl.edu** email. Approval is not always immediate, so apply early.
2. In VS Code, open the Copilot menu and choose **Use AI Features** or the sign-in option shown. Sign in with GitHub and finish the authorization in your browser. Follow any prompts to install extensions.
3. Open the Chat view, select a code cell in the notebook, and ask Copilot to explain it.

Full instructions: code.visualstudio.com/docs/setup/copilot. Whatever an AI assistant suggests, read the code and check the result yourself.